

The Space & Upper Atmosphere Research Commission, commonly referred to as SUPARCO, is the national space agency of Pakistan.

The agency, originally established in 1961 as a committee in Karachi, became an independent commission in 1981. Its initial objective was to study rocketry and high-altitude research from the United States, which eventually led to the development of a national satellite program. This culminated in the successful launch of Pakistan's first satellite from China in 1990. The agency was also an early participant in the rocket development program launched by the Ministry of Defence of Pakistan.

The agency leads the National Space Program (NSP) and maintains the orbital operations of its satellites with support facilities throughout the country.

SUPARCO also serves as the secretariat for Pakistan Space Activities Regulatory Board (PSARB) which was established in February 2024 under the National Command Authority to regulate all space-related activities in Pakistan.

History

Creation

The past federal ministries of Pakistan initially avoided to fund the space program and engineering education in spite of opportunity available from the United States. The Punjab University was the only university that was undertaking the research in aeronautics in 1957; after the former Soviet Union launched its first satellite in space, the Sputnik 1.

It was during the development of Apollo program in 1961 when Abdus Salam found an opportunity for Pakistan to start its space program with the foreign funding coming from the United States. As NASA launched the Apollo program in competition with the Soviet space program, it recognized the need for scientific data from the upper atmosphere and therefore invited India and Pakistan—two Indian Ocean–bordering states—to participate in related studies and experiments. Initially, the engineers from Pakistan Atomic Energy Commission

(PAEC) learned rocketry at the Wallops Flight Facility as Abdus Salam worked on getting approval for establishing a commission from the Ayub administration.

A commission to study the upper atmosphere and rocketry was established under Abdus Salam and Ishrat Usmani as its chairman with nuclear engineers from PAEC, Tariq Mustafa and Salim Mehmud, becoming its first members in 1961 through the "Space Sciences Research Wing" of PAEC. The Commission was the first in the Muslim world to start studies in establishing a national space program, and was named as "Space and Upper Atmosphere Research Commission" to represent its purpose and mission on 16 September 1961.

The Commission working under Abdus Salam was tasked with learning the rocket engineering, and contribution from France and the United States helped start this mission. Foundation of the agency made Pakistan the first South Asian country to start a space program. Furthermore, a Flight Test Range was established in Sonmiani which is 50 km (31 mi) west of Karachi, from where a program of sounding rocket launches was conducted based on the Nike-Ajax rockets followed by the Judi-Dart program.

On 7 July 1962, the Commission launched the first rocket, known as "Rehbar-I", which reached the altitude of 80 mi (420,000 ft) in space. The United States publicly supported and hailed the program as the beginning of "a program of continuous cooperation in space research of mutual interest." Until 1972, the United States provided training on rocket engines at the Goddard Space Flight Center. The ground stations for satellite navigation were set up by the Commission in Karachi and Lahore in 1973, and were visited by the Apollo 17 astronauts. In 1973, the Islamabad Ionospheric Station was established at the Quaid-e-Azam University and Landsat ground station was established near Lahore.

Funding and Support

Following the cessation of U.S. funding for upper atmosphere research in 1972 and the prioritization of nuclear weapons programs after the 1971 war with India, SUPARCO

experienced a significant decline in funding and support. Engineers with backgrounds in nuclear engineering were transferred to the Pakistan Atomic Energy Commission, and the Pakistan Air Force's support for the Rehbar program also ceased.

Following India's launch of its first satellite Aryabhata on a Soviet Kosmos-3M rocket in 1975, SUPARCO began lobbying for Pakistan's own satellite program. In 1981, the Commission was reorganized as an independent federal agency. During the same period, a communication satellite project called PakSAT was initiated. Collaborations with the Pakistan Radio Society (PRS) and the University of Surrey in England enabled SUPARCO to participate in satellite engineering projects, contributing to the development of UoSAT-1 and UO-11, which were launched in 1984.

Pakistan's communication satellite program also led to the expansion of a ground station in Lahore in 1983. SUPARCO began constructing its first satellite, Badr-1, in 1983, and it was eventually launched by China in 1990 after negotiations with the United States failed.

SUPARCO continued its collaboration with the United Kingdom, developing the Badr-B satellite in partnership with the British Rutherford Laboratory. However, due to orbital crises and funding constraints, Badr-B was not launched until 2001 by Russia.

Gen Zia's decision to delay the PakSAT project in 1984, citing a lack of funds, led to a significant setback for SUPARCO. This ultimately resulted in the loss of two orbital slots in 1993 and 1994. To secure a priority slot, the agency negotiated with Hughes Satellite Systems to acquire PakSAT-1, a geo-stationary satellite originally intended for Indonesia.

Functions

As per the National Space Policy of Pakistan approved in 2024, SUPARCO as the National Space Agency is mandated to carry out all activities related to outer space which includes but not limited to the following:

Policy Development: Formulate national space policies and legislation to comply with international obligations and establish guidelines for the space sector.

Program Management: Plan, manage, and execute the National Space Program, encompassing space science, technology, and applications.

Space Infrastructure: Design, develop, launch, and operate satellites, ground control infrastructure, space transportation systems, launch facilities, navigation systems, and tracking observatories.

National Register: Maintain a register of space objects launched by Pakistan and submit information to the United Nations.

Commercialization: Promote the commercial exploitation of space capabilities, technologies, and applications.

Private Sector Engagement: Encourage private sector involvement in space activities.

International Cooperation: Coordinate with international space organizations and agencies, and represent Pakistan in relevant forums.

Programs

Satellite Programs

Badr

PakSAT

The PakSAT program is the national communication satellites program of the commission conceived in 1979–80. The program is envisioned to consist of two geostationary communication satellites – one operating in 38°E orbit and other at 41°E, respectively.

The PakSAT program was originally designed to develop the television receive-only (TVRO) terminals for the receptions of news, entertainment, and educational channels from direct broadcasting satellite dishes.

Remote Sensing Satellites

CubeSats

Rocket Programs

Sounding Rockets

Since 1961, the commission supported and led the early studies on solid-propellant rockets, which it succeeded in developing the Rehbar-I. The Rehbar-I rocket was a derivative based on the U.S. Nike-Cajun, and continued its service until 1972.

Hatf-I & Abdali

In 1987, the military funded the commission's design study on rocket engines for Hatf-I, which was completed with the Khan Research Laboratories (KRL), the national defense laboratory of the Ministry of Defence. In 1995, the Commission designed the rocket engine for the Abdali project, which was completed in 2004.

The Commission also conducted studies on rocket engines for the Shaheen program.

Space Science and Astronomy

Every year, SUPARCO sponsors and organizes the World Space Week (WSW) to promote the understanding of the Earth science all over the country. SUPARCO works with a number of universities and research institutions to engage in research in observational astronomy and astrophysics. The Institute of Space and Planetary Astrophysics (ISPA) of the Karachi University conducts key research and co-sponsors with international level research programs in astrophysics, with joint ventures of SUPARCO.

SUPARCO operates a national balloon launching facility in Karachi to conduct studies in atmospheric sciences to determine the vertical profile of ozone up to 30–35 km. This balloon sounding facility has been extensively used for carrying out research in better understanding of the meteorology and how the ozone layer vary seasonally in the stratosphere and

troposphere. The Ionospheric Station at Karachi operates a Ionosonde observation facility, and recently the balloon flight mission was carried out by the station on 16 January 2004, up to an altitude of about 36 km to measure the vertical profile of the O₃ trends. The maximum O₃ observed 12.65 mPa at 27 km. One of the most notable mission of SUPARCO is its Lunar program that conducts observational studies on the activity of Lunar phases and distributes its publications within the public domain.

The SUPARCO Astrophysics program, is an active scientific mission of SUPARCO, dedicated for the development of space science. The program's mainstream objective and aim is to conduct research studies for the advancement and better understanding of the theoretical physics, astronomy, astrophysics, and mathematics involving the three-dimensional universal space and time.

SUPARCO's Space Program 2040, launched in 2012, incorporates astronomy and astrophysics research into a single program focused on theoretical and observational studies. This program explores vast topics like quantum mechanics, deep space objects, dark matter and energy, supernovae, nebulae, and galaxies. Aligned with Pakistan's official space policy, it also aims to strengthen public understanding of physics and mathematics through educational initiatives like academic bulletins and public events celebrating astronomy milestones. This program fosters collaboration with international space agencies and builds upon public interest sparked by SUPARCO's celebration of the International Year of Astronomy in 2009 which was widely appreciated by the public.

Since its establishment, a total of nine important publications has been released under the auspicious of this program with the last volume was issued in September 2012.

Space Weather Monitoring

SUPARCO has been actively involved in space weather monitoring for over five decades, utilizing a nationwide network of ground-based sensors. As Pakistan's reliance on satellites for

critical services like communication, navigation, and earth observation increases, SUPARCO has established the Pakistan Space Weather Center, which employs an array of instruments to observe space weather phenomena in real time. Processed data, including HF communication products, are then disseminated to relevant national users.

SUPARCO initiated its Geomagnetic Field Monitoring Program in 1983 at the Sonmiani space facility, establishing a second observatory in Karachi in 2008. This program involves studying the Earth's magnetic field and its variations in the South Asian region using data collected from these observatories. The data is used to understand the Earth's magnetic environment and mitigate associated hazards.

SUPARCO regularly publishes a public domain bulletin of geomagnetic data, which includes research on the effects of solar flares and severe magnetic storms recorded by the observatories.

On Sunday, 19th October 2025, SUPARCO announced the "successful launch" of their new H1 satellite from the Jiuquan Satellite Launch Centre in north-western China.

This is Pakistan's first hyperspectral satellite, described as a "major milestone". It aims to advance country objectives ranging from agriculture to urban planning.

Pakistan's Foreign Ministry says the new technology will greatly improve the country's abilities in areas like advanced farming, environmental protection, city development, and disaster response. It will also help identify natural hazard risks more accurately, supporting major development projects such as the China-Pakistan Economic Corridor.

Pakistan praised the launch of H1 as a major milestone for its space programme and a sign of its close cooperation with China in using space technology for peaceful purposes.

Satellite Navigation Program

SUPARCO has been a pivotal player in Pakistan's satellite navigation landscape. Its involvement extends to the entire spectrum of satellite navigation, encompassing the design

and development of space, ground, and user segments.

It has established a Ground Based Augmentation System (GBAS) on a proof of concept basis to provide correction signals to authorized users. Additionally, SUPARCO has deployed a Space Based Augmentation System (SBAS) via PakSAT-MM1 to cater to the specific needs of aviation, marine, and land users who require high-integrity correction signals.

SUPARCO operates a Satellite Navigation Signal Monitoring facility that plays a crucial role in monitoring, archiving, and analyzing satellite navigation signals from various monitoring stations located across Pakistan. This facility fosters collaboration with national and international organizations involved in satellite navigation systems.

International COSPAS-SARSAT Programme

Pakistan has been participating in a multinational humanitarian programme for satellite-aided search and rescue, the International Cospas-Sarsat Programme. In 1990, the Government of Pakistan accorded approval for SUPARCO's participation in the Cospas program as ground segment provider and lead space station with close coordination with the Soviet Union. Over the years, the mission control center has equipped itself with advanced technology and is capable of distributing distress alert data to rescue coordination centres in the country.

Remote Sensing Program

SUPARCO has been a pioneer in introducing Remote Sensing/GIS and allied technologies in Pakistan, providing turnkey solutions and services to diverse users across various fields. These applications range from agriculture and forestry to disaster management, water resources, environmental monitoring, urban planning, and coastal and marine studies. SUPARCO's expertise extends to climate change and environmental degradation, utilizing ground-based observations and satellite data for now-casting and forecasting environmental indicators in

the atmosphere, biosphere, cryosphere, and hydrosphere. To support these endeavors, SUPARCO operates specialized centers and facilities, including environmental laboratories, mobile laboratories, a mathematical modeling center, and a microgravity experiments facility.

Other Scientific programs

Scientific space research

Geographic Information Systems

Natural Resource Surveying

Environmental monitoring

Acquisition of data for atmospheric/meteorological studies

Development of the ground-based infrastructure for navigation and special information system

Development of research, test and production base of the space sector

Facilities

SUPARCO, headquartered in Islamabad, maintains a network of key technical and support facilities across Pakistan. These facilities are strategically located in major cities to support the organization's various research and development activities.

Ground Stations

Space Applications Centres

Research and Development Centres

Space Ports

Human resource development

International Cooperation

SUPARCO, has consistently prioritized international cooperation as a central component of its

space development program. The agency maintains active membership in several international organizations, institutes, scientific committees and United Nations bodies like UNCOPUOS, UN-SPIDER, UN-ESCAP, COSPAR, IAF, ISPRS, APSCO etc. This has fostered collaboration in scientific and technical information exchange, data sharing, joint projects, technology transfer, training, collaborative studies. SUPARCO has also entered into numerous bilateral and multilateral cooperation agreements and MoUs to facilitate space-related activities. Pakistan is a signatory to all five United Nations treaties governing the peaceful uses of outer space.

Inter-Islamic Network on Space Sciences and Technology

ISNET was established in 1987 by the Organization of Islamic Cooperation (OIC) Standing Committee on Scientific and Technological Cooperation (COMSTECH) and SUPARCO hosted the founding meeting of the Network at Karachi. The secretariat of ISNET is located within SUPARCO and Chairman SUPARCO is the ex-officio President of ISNET. ISNET has 17 member states and it organises different international conferences, seminars, workshops, training courses related to space science, technology and satellite application in different member states in which a number of scientists, engineers and researchers stake-holders participate.

China

In August 2006, People's Republic of China signed an agreement with Pakistan to conduct joint research in space technology and committed to work with Pakistan to launch three Earth-weather satellites over the next five years. In May 2007, China (as a strategic partner) publicly signed an agreement with Pakistan to enhance cooperation in the areas of space science and technology. The Pakistan-China bilateral cooperation in the space industry span a broad spectrum, including climate science, clean energy technologies, atmospheric and Earth sciences, and marine sciences. On the occasion of Chinese launch of PakSAT-1R, Pakistan's

ambassador to China expressed the natural desire of Pakistan for China to send the first officially designated Pakistani astronaut to space aboard a Chinese spacecraft. In cooperation with CNSA, Pakistan sent its first lunar orbiter mission called ICUBE-Q along with Chang'e 6. SUPARCO will also launch a 35kg lunar rover with the Chang'e 8 mission.

In April 2026, the China Manned Space Agency (CMSA) announced that Muhammad Zeeshan Ali and Khurram Daud has been selected as its first two international astronaut candidates. One of the two is expected to fly to the Chinese Tiangong space station for a one-week visit on the Shenzhou 24 mission in October 2026.

United Arab Emirates

In 2019, the commission reached out to United Arab Emirates Space Agency to take part in the Global Space Congress for the first time held at Abu Dhabi, where they held an exhibition on their satellite-related projects.

Turkey

In December 2006, Turkey showed interest to form a joint-venture with Asia-Pacific Space Cooperation Organization where Pakistan is a member. In 2006, Turkish minister of science, accompanied by the Turkish Ambassador to Pakistan, signed the Memorandum of understanding (MOU) with Pakistan to form a joint-venture with Pakistan in the development of satellite technology. The Scientific and Technological Research Council of Turkey and Turkish Aerospace Industries's senior ranking officials and representative signed a separate accord with the SUPARCO to enhance the cooperation in the satellite development program.

Leadership History

See also

List of government space agencies

SUPARCO's spaceflight missions and tests

SUPARCO Space Programme 2040

Jinnah Antarctic Station

Pakistani missile research and development program

Notes

References

Sources

External links

CNBC Pakistan televised Interview with Salim Mehmud – Chairman SUPARCO - Available in Urdu language

Space and Upper Atmosphere Research Commission (SUPARCO) Archived 30 July 2020 at the Wayback Machine

FAS report on SUPARCO Archived 11 July 2016 at the Wayback Machine

Astronautix

Shaheen - PSLV